

CHAPTER 3

Network Devices

NIC and MAC address · modem · repeater · hub · bridge · switch · router · gateway · access point · collision and broadcast domains

QUICK TRICK

Learn the fast tricks first (details below):

1. **Layer rule:** Repeater and Hub = Layer 1 · Bridge and Switch = Layer 2 · Router = Layer 3 · Gateway = upper layers.
2. **Hub** repeats to **all** ports. **Switch** sends only to the **right** port (uses MAC address). **Router** sends between networks (uses IP address).
3. **Modem** = digital ↔ analog. **NIC** = card that gives the device a **48-bit MAC address**.
4. **Collision domains:** hub = 1 for all ports; switch / bridge = 1 per port. **Broadcast domains:** router = 1 per interface; switch / hub do not split them.

3.1 What are network devices?

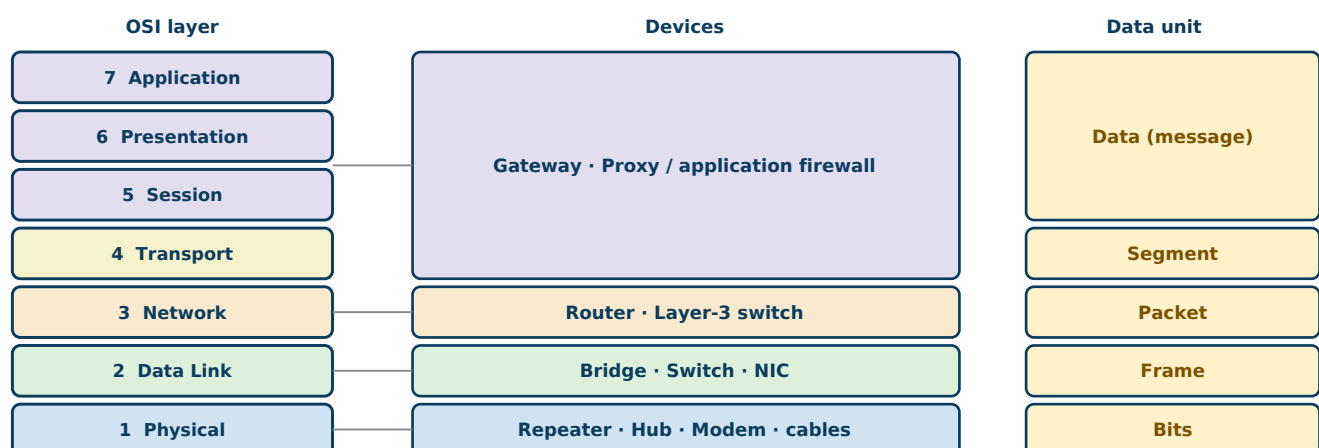
● LEVEL 1 · BEGINNER

The problem. Cables alone cannot build a network. A cable has no brain: it cannot decide **where** data should go, clean up a weak signal, or join two different networks. **The idea.** Special **network devices** do these jobs. Some only boost signals; some choose the right path; some convert one form of data into another.

Analogy. A city. Roads are cables. A **repeater** is a loudspeaker on a long road. A **hub** is a village crier who shouts every message to everyone. A **switch** is a postman who knows every house and delivers to the right door. A **router** is the post office that sends letters to other cities. A **gateway** is a translator at the border.

Kind of device	Meaning	Examples
End device (host)	Source or destination of data	PC, laptop, phone, printer, server
Intermediate device	Moves, filters or converts data between hosts	Repeater, hub, bridge, switch, router, gateway, modem, access point

Devices are grouped by the **OSI layer** where they work. The OSI model is explained fully in a later chapter; for now remember that a **lower layer = simpler device** (only bits), and a **higher layer = smarter device** (understands addresses and protocols).



QUICK TRICK

Bottom-up memory: **R-H** (Repeater, Hub) at layer 1 → **B-S** (Bridge, Switch) at layer 2 → **R** (Router) at layer 3 → **G** (Gateway) at the top. The address used also changes: none → MAC → IP.

3.2 NIC and MAC address

● LEVEL 1 • BEGINNER

A **Network Interface Card (NIC)** is the hardware that connects a computer to a network. It has the port (RJ-45 for Ethernet or an antenna for Wi-Fi), converts data to signals, and gives the device its unique **MAC address**. Names you may see: Ethernet card, LAN card, network adapter, **WNIC** (wireless NIC, Wi-Fi card). Today a NIC is usually built into the motherboard.

MAC address (Media Access Control address) is the **physical or hardware address** of a NIC. It is **48 bits (6 bytes)** long and is written as 12 hexadecimal digits, for example **00:1A:2B:3C:4D:5E** (also written 00-1A-2B-3C-4D-5E). It is fixed by the manufacturer (burned into the card), and it works at the **Data Link layer** (Layer 2).

Part	Size	Meaning	In 00:1A:2B:3C:4D:5E
OUI (Organizationally Unique Identifier)	First 24 bits (3 bytes)	Manufacturer code given by IEEE	00:1A:2B
Device serial part	Last 24 bits (3 bytes)	Number chosen by the manufacturer for each card	3C:4D:5E

	MAC address	IP address
Also called	Physical / hardware address	Logical address
Size	48 bits (12 hex digits)	32 bits (IPv4) or 128 bits (IPv6)
Given by	Manufacturer (fixed in the card)	Network administrator / DHCP (can change)
Works at	Layer 2 (Data Link)	Layer 3 (Network)
Scope	Within one local network (one hop)	Across networks (end to end)
Used by	Switch, bridge	Router

Special MAC addresses: **FF:FF:FF:FF:FF:FF** is the **broadcast** address (every device on the LAN accepts the frame). A frame sent to one specific MAC is **unicast**. Addresses starting with 01:00:5E are used for IPv4 **multicast**.

How many MAC addresses exist? $2^{48} = 281\,474\,976\,710\,656$, about 2.8×10^{14} .

COMMON MISTAKE

A MAC address is **not** the same as an IP address. MAC identifies the **card** inside one LAN; IP identifies the **host on the network** and changes when you move to another network. (A MAC can be changed by software, called spoofing, but its factory value is fixed.)

3.3 Modem

● LEVEL 1 • BEGINNER

Modem = **MO**dulator + **DEM**odulator. It converts the **digital** signal of a computer into an **analog** signal that can travel on a telephone or cable line (**modulation**), and converts the incoming analog signal back into digital (**demodulation**).

Analogy. A modem is an interpreter between two people who speak different languages: the computer speaks digital (0 and 1) and the old telephone line speaks analog (waves).

Type	Line used	Note
Dial-up modem	Ordinary telephone line	Up to 56 kbps; line is busy while online
DSL / ADSL modem	Telephone copper line	ADSL = Asymmetric: download speed is higher than upload. Voice (up to 4 kHz) and data use different frequencies, so you can talk and browse together.
Cable modem	Coaxial TV cable	Uses DOCSIS standard; shared by neighbourhood
Fibre terminal (ONT)	Optical fibre	Converts light to electrical Ethernet signals (FTTH broadband)
Mobile / USB dongle	Cellular network (4G / 5G)	Wireless modem in a small stick or phone

Basic modulation methods

Method	What changes	Short name
Amplitude Shift Keying	Height (amplitude) of the carrier wave	ASK
Frequency Shift Keying	Frequency of the carrier wave	FSK
Phase Shift Keying	Phase (starting angle) of the carrier wave	PSK

QUICK TRICK

Modem connects you to the ISP; router shares that connection with your devices. At home both are often inside one box (a "Wi-Fi modem router").

3.4 Repeater

● LEVEL 1 · BEGINNER

Repeater is a Layer 1 device that **receives a weak signal, regenerates it (cleans and re-times it) and sends it on**. It is used to extend the distance of a cable, because a signal loses strength (attenuation) as it travels. It has two ports and does not look at addresses; it repeats every bit, even noise-free errors and collisions.

Point	Repeater	Amplifier
Signal type	Digital (regenerates exact 0s and 1s)	Analog
Noise	Removes noise, makes a fresh signal	Amplifies signal and noise together
Layer	Physical (1)	—

BPSC POINT

A repeater joins two segments of the **same** network and the **same** protocol. It cannot filter traffic or connect different networks.

5-4-3 rule (old 10 Mbps Ethernet with repeaters): at most **5 segments**, joined by **4 repeaters**, and only **3 segments** may have computers on them. With 10BASE5 (500 m) this gives a maximum of $5 \times 500 = 2500$ m.

A wireless **range extender** works like a repeater for Wi-Fi.

3.5 Hub

● LEVEL 1 · BEGINNER

Hub is a multi-port repeater (Layer 1). A signal that comes in on one port is **copied to all other ports**. Every computer receives every frame, and only the one with the matching address keeps it. The whole hub is **one collision domain**, so two computers sending together cause a **collision**. Hubs work in **half duplex** and the bandwidth is **shared**.

Type	Meaning
Passive hub	Only a connection point; no power, no signal boost
Active hub	Needs power; regenerates the signal like a repeater
Intelligent (smart) hub	Active hub with management features such as monitoring

Disadvantages: collisions grow with more users; shared bandwidth (a 100 Mbps hub with 5 busy PCs gives about $100 \div 5 = 20$ Mbps each); anyone can capture the traffic of all (poor security); no filtering. Hubs are now **obsolete** and replaced by switches.

3.6 Bridge

● LEVEL 2 · EXAM

Bridge is a Layer 2 device that joins **two LAN segments** and forwards a frame only when needed. It reads the **MAC address** of each frame and keeps a **bridge (forwarding) table** of which MAC address is on which side. It therefore **filters** traffic: a frame meant for a computer on the same side is not sent across.

Step	What the (transparent) bridge does
1. Learn	Reads the source MAC of an incoming frame and stores it with the port it came from
2. Look up	Looks for the destination MAC in the table
3. Forward	If the destination is on another port, send only to that port
4. Filter	If the destination is on the same port, drop the frame
5. Flood	If the destination is unknown (or broadcast), send to all other ports

Types: **transparent** (learning) bridge used in Ethernet; **source-routing** bridge used in Token Ring; **translational** bridge joins different LAN types (for example Ethernet to Token Ring). Bridges can form loops, so **Spanning Tree Protocol (STP, IEEE 802.1D)** blocks the extra paths.

BPSC POINT

A bridge **splits collision domains** (each port is its own), but it **does not split the broadcast domain** (broadcasts are flooded everywhere). A bridge is slower than a switch because it does the work in software and has few ports (usually 2).

3.7 Switch

● LEVEL 1 · BEGINNER

Switch is a Layer 2 device with many ports (8, 24, 48 and more). It works like a **multi-port bridge** but uses fast hardware (ASIC). It learns the MAC address of each connected device in a **MAC address table** (also called CAM table), and sends a frame **only to the port of the destination**. Each port is its own collision domain, and ports can work in **full duplex**, so every pair of devices can talk at full speed at the same time.

How a switch learns (example)

PCs A, B and C are connected to ports 1, 2 and 3 of a new switch. The table is empty at the start.

Step	Frame	Table after the step	What the switch does
1	A → B	A = port 1	B unknown, so flood to ports 2 and 3
2	B → A	A = 1, B = 2	A known, so send only to port 1
3	A → B	A = 1, B = 2	B known, so send only to port 2
4	C → A	A = 1, B = 2, C = 3	A known, send only to port 1

Switching mode	When it forwards	Speed / error check
Store-and-forward	After the whole frame is received	Slowest; checks errors (CRC) and drops bad frames
Cut-through	After reading only the destination MAC (first 6 bytes of the frame header)	Fastest; no error check
Fragment-free	After the first 64 bytes	Middle; avoids collision fragments

Other terms: **Unmanaged** switch (plug and play) vs **managed** switch (can be configured, supports VLAN, monitoring). **VLAN** (Virtual LAN) splits one switch into separate logical networks, each a different broadcast domain. A **Layer-3 switch** can also route between VLANs using IP addresses.

Point	Hub	Bridge	Switch
Layer	1	2	2 (some 3)
Ports	Many	Usually 2	Many
Sends frame to	All ports	Needed port (by MAC)	Needed port (by MAC)
Collision domains	1 for the whole hub	1 per port	1 per port
Duplex	Half	Half / full	Full
Bandwidth	Shared	Per segment	Dedicated per port
Security	Poor	Better	Good
Status	Obsolete	Rare	Standard in every LAN

3.8 Router

● LEVEL 2 · EXAM

Router is a Layer 3 device that **connects different networks** and forwards **packets** using **IP addresses**. It checks the destination IP of each packet, looks it up in its **routing table**, and sends the packet through the best interface towards the next hop. A router joins your LAN to the WAN or Internet.

Routing table column	Meaning
Destination network	The network to reach (for example 192.168.1.0)
Subnet mask	Tells which part of the address is the network part
Next hop / gateway	The next router to send the packet to
Interface	The port through which to send
Metric	Cost of the path (hops, speed); the lowest is best

Static routing	Dynamic routing
Routes entered by the administrator by hand; no overhead; does not adapt to failures	Routers learn routes with protocols (RIP, OSPF, BGP) and update automatically after a failure

Functions of a router: path selection and forwarding, connecting different networks, **separating broadcast domains**, and in home routers also **NAT** (one public IP for many private devices), **DHCP** (gives IP addresses automatically), a basic **firewall** and Wi-Fi. Routing protocols and IP addressing are covered in later chapters.

Point	Switch	Router
Layer	2	3
Address used	MAC	IP
Connects	Devices in the same network	Different networks
Data unit	Frame	Packet
Broadcast	Forwards broadcasts (one domain)	Blocks broadcasts (each interface a domain)
Table	MAC address table	Routing table

QUICK TRICK

Think of the **default gateway** in your computer settings: it is simply the **IP address of the router** that leads out of your LAN.

3.9 Gateway

● LEVEL 2 · EXAM

Gateway is a device (or software) that connects **two networks that use different protocols or architectures** and **converts** data between them. It can work at any layer, but mostly at the upper layers (up to Application). It is the most intelligent and slowest of the connecting devices.

Example	What it converts
E-mail gateway	Between different mail systems or formats
VoIP gateway	Telephone (PSTN) voice ↔ Internet packets
Protocol gateway	Between different protocol suites, for example TCP/IP ↔ an older proprietary network
Default gateway	In daily use: the router that gives a LAN access to other networks

COMMON MISTAKE

Do not confuse the **gateway** (protocol converter, upper layers) with the **default gateway** (just the router's IP address). Exam options often use the word loosely; read the question for "convert protocols" (gateway) or "leave the network" (router).

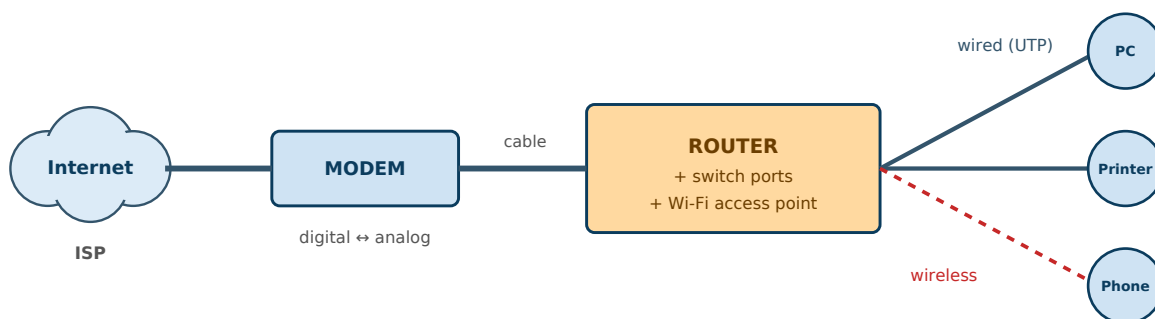
3.10 Wireless access point and Wi-Fi

● LEVEL 1 • BEGINNER

Access Point (AP) connects wireless devices to a wired network using **Wi-Fi (IEEE 802.11)**. It works at Layer 2. It broadcasts a network name called the **SSID**. A **wireless router** is a router + switch + access point in one box. Wi-Fi uses the 2.4 GHz, 5 GHz (and now 6 GHz) bands.

Standard	Marketing name	Band	Top speed (theory, about)
802.11b	—	2.4 GHz	11 Mbps
802.11a	—	5 GHz	54 Mbps
802.11g	—	2.4 GHz	54 Mbps
802.11n	Wi-Fi 4	2.4 and 5 GHz	600 Mbps
802.11ac	Wi-Fi 5	5 GHz	About 6.9 Gbps
802.11ax	Wi-Fi 6 / 6E	2.4, 5 (and 6 GHz in 6E)	About 9.6 Gbps
802.11be	Wi-Fi 7	2.4, 5 and 6 GHz	About 46 Gbps

A typical home network



Other devices and tools you may be asked about

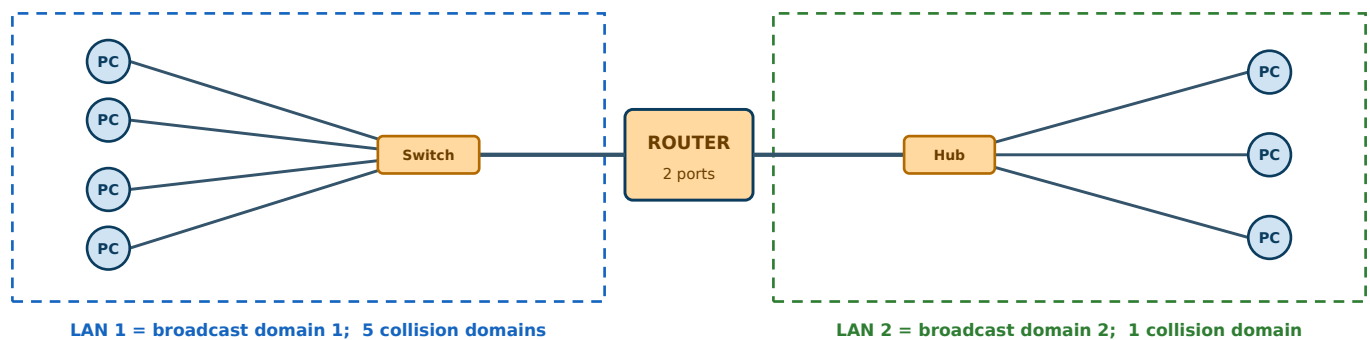
Device / tool	Purpose
Firewall	Filters traffic by rules to block unwanted access (hardware or software); covered in the security chapter
Proxy server	Middle server that fetches pages on behalf of clients; caching and filtering
Media converter	Converts between media, for example copper UTP ↔ optical fibre
Patch panel	Board where cables end, so connections can be changed easily with short patch cords
Brouter	Bridge + router in one device
Crimping tool	Fixes an RJ-45 connector on a UTP cable
LAN (cable) tester	Checks that all 8 wires of a cable are connected properly

3.11 Collision domain and broadcast domain

● LEVEL 3 • BPSC / PGT

Collision domain: the part of a network where two devices sending at the same time will collide. **Broadcast domain:** the part of a network where a broadcast frame sent by one device is received by all the others.

Device	Effect on collision domains	Effect on broadcast domains
Repeater / hub	Does not split (all ports = one domain)	Does not split
Bridge / switch	Splits: each port = one domain	Does not split (unless VLANs are used)
Router	Splits: each port = one domain	Splits: each interface = one domain



Worked example. A router has two interfaces. LAN 1: a switch with 4 PCs, joined to the router. LAN 2: a hub with 3 PCs, joined to the router.

Collision domains: LAN 1 has 4 PC links + 1 switch-to-router link = 5 (a switch gives one domain per port). LAN 2 (hub + its 3 PCs + the router port) is 1. Total = **6**.

Broadcast domains: one per router interface = **2**.

QUICK TRICK

Counting recipe: **(1)** every switch or bridge **port that is used** is one collision domain; **(2)** a hub and everything on it, up to the switch or router port, is **one** collision domain; **(3)** broadcast domains = number of router interfaces in use (with no VLANs).

3.12 All devices at a glance

● LEVEL 2 · EXAM

Device	OSI layer	Address used	Main job	Collision domains	Broadcast domains
Repeater	1	None	Regenerate signal, extend distance	Same	Same
Hub	1	None	Copy signal to all ports	One for all	One for all
Modem	1 / 2	—	Digital ↔ analog conversion	—	—
NIC	1 / 2	MAC	Connect host; carries MAC address	—	—
Bridge	2	MAC	Join two LANs; filter	1 per port	Same
Switch	2	MAC	Send frame to right port	1 per port	Same (VLAN splits)
Access point	2	MAC	Wireless to wired link	Shared radio	Same
Router	3	IP	Connect networks, path selection	1 per port	1 per interface
Gateway	Upper (4–7)	IP / app	Convert protocols	—	—

3.13 Common mistakes

● LEVEL 2 · EXAM

COMMON MISTAKE

- Saying a switch works on IP addresses. A normal switch uses **MAC**; only a Layer-3 switch or router uses IP.
- Saying a hub is smart. A hub does **no filtering**; it repeats to all ports.
- Thinking a switch splits broadcast domains. It does **not**; a **router** (or VLAN) does.
- Treating a repeater as an amplifier. A digital repeater **regenerates**; an amplifier also boosts noise.
- Mixing modem and router. Modem = reach the ISP; router = share and route.
- Using "gateway" and "default gateway" as the same thing.
- Counting hub ports as separate collision domains. All hub ports are **one** domain.

Quick-Solve Tricks — Summary

Situation	Trick
Layer 1 devices	Repeater, hub (and cables)
Layer 2 devices	Bridge, switch, NIC, access point
Layer 3 device	Router (IP address)
Protocol conversion	Gateway
Digital ↔ analog	Modem
Send to all ports	Hub (and a switch for broadcast or unknown MAC)
MAC address	48 bits, 12 hex digits; first 24 bits = OUI, last 24 = serial
Broadcast MAC	FF:FF:FF:FF:FF:FF
Collision domains	Hub = 1 total · switch/bridge/router = 1 per port
Broadcast domains	Router = 1 per interface; switch and hub do not split
Switch learning	Learns source MAC, forwards by destination MAC, floods if unknown
5-4-3 rule	5 segments, 4 repeaters, 3 populated segments
Default gateway	IP address of the router out of the LAN

BPSC QUICK CHECK — Chapter 3

NIC → MAC address (48 bits)	Modem → modulator + demodulator
Repeater → regenerates the signal	Hub → multi-port repeater, half duplex
Bridge → filters by MAC, 2 ports	Switch → multi-port bridge, full duplex
Router → IP address, connects networks	Gateway → converts protocols
STP (802.1D) → stops bridge loops	Wi-Fi → IEEE 802.11, SSID, access point

Traps: MAC vs IP · hub vs switch · switch vs router for broadcasts · gateway vs default gateway · amplifier vs repeater · collision domain vs broadcast domain.

Practice questions

● LEVEL 1 · BEGINNER

- **Q1.** Expand NIC and MAC.
- **Q2.** Which device converts digital signals to analog and back?
- **Q3.** Which device connects a LAN to the Internet and uses IP addresses?
- **Q4.** Name two devices that work at the physical layer.
- **Q5.** How many bits are there in a MAC address?
- **Q6.** Which device sends the incoming data to all of its ports?

● LEVEL 2 · EXAM

- **Q7.** Give three differences between a hub and a switch.
- **Q8.** What does a bridge do when it receives a frame whose destination MAC is not in its table?
- **Q9.** How many collision domains are there in (a) a switch with 6 PCs, (b) a hub with 6 PCs?
- **Q10.** What is a default gateway?
- **Q11.** Write the OUI part of the MAC address 00:1A:2B:3C:4D:5E.
- **Q12.** Why is a digital repeater better than an amplifier?

● LEVEL 3 · BPSC / PGT

- **Q13.** One router has 3 interfaces. LAN 1: a switch with 5 PCs. LAN 2: a switch with 4 PCs. LAN 3: a hub with 6 PCs. Find the number of collision domains and broadcast domains.
- **Q14.** How many different MAC addresses can 48 bits give?
- **Q15.** A 100 Mbps hub has 5 PCs, all sending at the same time and sharing equally. What is the average speed per PC? What would each PC get on a full-duplex 100 Mbps switch?
- **Q16.** A new switch has an empty table. PC A (port 1) sends a frame to PC D (port 4). Then D sends a reply to A. Describe what the switch does each time.

- **Q17.** A 10BASE5 network follows the 5-4-3 rule. How many repeaters and segments can there be at most, and what is the maximum length?
- **Q18.** Match each to a layer: hub, switch, router, repeater, bridge.

Answers

Q	Solution
1	NIC = Network Interface Card. MAC = Media Access Control.
2	Modem.
3	Router.
4	Repeater and hub (also cables; a modem works at layers 1 / 2).
5	48 bits (6 bytes, 12 hexadecimal digits).
6	Hub.
7	Hub: Layer 1, sends to all ports, one collision domain, half duplex, shared bandwidth, no filtering. Switch: Layer 2, sends only to the destination port using the MAC table, one collision domain per port, full duplex, dedicated bandwidth.
8	It floods the frame to all ports except the one it came from (and learns the source MAC).
9	(a) 6 , one per switch port. (b) 1 , the whole hub is one domain.
10	The IP address of the router that a host uses to reach devices outside its own network.
11	00:1A:2B (first 24 bits, the manufacturer code).
12	A repeater regenerates a clean digital signal, so noise is removed. An amplifier also amplifies the noise.
13	Collision domains: LAN 1 = 5 PCs + 1 uplink = 6; LAN 2 = 4 PCs + 1 uplink = 5; LAN 3 (hub + router port) = 1. Total = 12 . Broadcast domains = 3 (one per router interface).
14	$2^{48} = 281\,474\,976\,710\,656 \approx 2.8 \times 10^{14}$.
15	Hub: $100 \div 5 = 20$ Mbps each on average. Full-duplex switch: 100 Mbps in each direction per PC, since each port has its own link.
16	A → D: table empty, so the switch learns A = port 1 and floods to ports 2, 3, 4 (D is unknown). D → A: learns D = port 4, finds A on port 1 and sends it only to port 1 . Now both are known.
17	At most 5 segments and 4 repeaters , with only 3 segments holding computers. Maximum length = $5 \times 500 \text{ m} = 2500 \text{ m}$.
18	Hub: 1. Repeater: 1. Switch: 2. Bridge: 2. Router: 3.

REMEMBER

Next: Chapter 4 — Switching, Multiplexing and Error Detection (circuit, message and packet switching; FDM, TDM, WDM; parity, checksum, CRC; framing).

Roadmap: Ch 5 OSI and TCP/IP Models · Ch 6 Network Protocols (HTTP, FTP, SMTP, POP3, TELNET, PPP, VoIP, DNS) · Ch 7 IP Addressing and Subnetting · Ch 8 Web Services · Ch 9 Network Security · Ch 10 Mobile Telecom, Cloud and IoT.